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10.6 Surface Area and Volume Under Dilation

When a solid is dilated (scaled) by factor k, every *linear* measurement (edges, radii, heights, slant heights) scales by $k^1 = k$. Surface area is built from products of two lengths (length × width), so it scales by k^2; volume is built from products of three lengths, so it scales by k^3. This is why doubling a solid's size ($k = 2$) quadruples its surface area but multiplies its volume by 8 — a fact that shows up everywhere from packaging costs to biology (why small animals lose heat faster than large ones, since SA scales slower than volume). This worksheet computes new SA/volume after a dilation, recovers k from a given ratio, and converts between SA ratios and volume ratios.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. SA $= 50$, $k = 3$. Find the new SA.

2. Volume $= 12$, $k = 2$. Find the new volume.

3. SA ratio $16 : 1$. Find k.

4. Volume ratio $125 : 1$. Find k.

5. SA $= 180$, $k = \frac{1}{3}$. Find the new SA.

6. Volume $= 64\pi$, $k = \frac{1}{2}$. Find the new volume.

7. $k = 5$. Find the SA ratio (new:old).

8. $k = \frac{2}{3}$. Find the volume ratio (new:old).

9. Volume ratio $8 : 27$. Find the SA ratio.

10. Edge 10, $k = 0.4$. Find the new edge.

11. SA $= 72$, $V = 48$, $k = 2$. Find the new SA and new V.

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12. SA ratio $4:9$ (small:large). Find k (small to large).

13. Lengths scale by k. State how SA and volume scale.

14. A small object has SA $= 15$ and volume $= 10$. A similar large object has volume $= 80$. Find k and the large object's SA.

15. Two similar cones have volumes in ratio $\frac{27}{8}$ (large:small). The small cone has slant height 10 and total SA 75π. Find (a) k, (b) the large cone's slant height, (c) the large cone's SA.

✓ Answer Key

1. 450
2. 96
3. 4
4. 5
5. 20
6. 8π
7. $25:1$
8. $8:27$
9. $4:9$
10. 4
11. SA $= 288$; $V = 384$
12. $\frac{3}{2}$
13. SA by k^2; volume by k^3
14. $k = 2$; SA $= 60$
15. $k = \frac{3}{2}$; slant height 15; SA $= \frac{675\pi}{4}$